

UNIT I – INTRODUCTION

1. What is Ethical Hacking?

Ethical hacking is the legal process of testing computers, networks, or websites to find security weaknesses.

Example: A company hires an ethical hacker to test whether its website is secure.

2. Define Ethical Hacker

An ethical hacker is a security expert who uses hacking methods legally to identify vulnerabilities in systems and networks.

3. What is Penetration Testing?

Penetration testing is a security testing process where ethical hackers simulate attacks to identify vulnerabilities.

Example: Testing whether weak passwords can be hacked.

4. What are Penetration Testing Methodologies?

Penetration testing methodologies are step-by-step procedures followed during security testing.

Main Steps:

1. Information gathering
 2. Scanning
 3. Gaining access
 4. Maintaining access
 5. Reporting
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5. What is TCP/IP?

TCP/IP stands for Transmission Control Protocol / Internet Protocol. It is the communication protocol used to transfer data over the Internet.

Functions:

- TCP ensures reliable communication
 - IP handles addressing and routing
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6. What is the Application Layer?

The application layer provides services directly to users and applications.

Examples:

- HTTP
- FTP
- SMTP

Uses: Web browsing, email, and file transfer.

7. What is the Transport Layer?

The transport layer manages end-to-end communication and reliable data transfer.

Protocols:

- TCP(Transmission Control Protocol)
- UDP(User Datagram Protocol)

Functions:

- Error checking
 - Data delivery
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8. What is the Internet Layer?

The internet layer handles logical addressing and routing of packets between networks.

Main Protocol: IP Protocol

9. What is IP Addressing?

IP addressing gives a unique address to every device connected to a network.

It helps devices communicate with each other.

Example: 192.168.1.1

10. What are Network Attacks?

Network attacks are malicious activities performed to damage or access networks illegally.

Examples:

- DoS attack
 - Packet sniffing
 - Spoofing
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11. What is Malware?

Malware is harmful software designed to damage systems or steal information.

Types:

- Virus
 - Worm
 - Trojan
 - Ransomware
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12. How to Protect Against Malware?

Protection Methods:

- Install antivirus software
- Update operating systems
- Avoid unknown links
- Use firewall protection

These methods help reduce malware attacks.

13. What is an Intruder Attack?

An intruder attack happens when an unauthorized user accesses a system illegally.

Purpose:

- Steal data
 - Damage systems
 - Spy on users
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14. What is Physical Security?

Physical security protects hardware and systems from theft, damage, or unauthorized access.

Examples:

- CCTV cameras
 - Door locks
 - Security guards
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15. What is Firewall?

A firewall is a security system that monitors and filters network traffic.

Purpose:

Blocks unauthorized access to networks.

16. Difference Between Ethical Hacker and Malicious Hacker

Ethical Hacker	Malicious Hacker
Works legally	Works illegally
Has permission	No permission
Protects systems	Damages systems

17. Difference Between TCP and UDP

TCP	UDP
Reliable	Faster
Connection-oriented	Connectionless
Error checking available	No error checking

UNIT II – FOOTPRINTING, RECONNAISSANCE AND SCANNING NETWORKS

1. What is Footprinting?

Footprinting is the process of collecting information about a target system or organization.

2. What is Reconnaissance?

Reconnaissance is the first stage of ethical hacking where information about the target is collected.

Types:

- Active reconnaissance
 - Passive reconnaissance
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3. What is Footprinting through Search Engines?

Search engines are used to collect publicly available information about a target.

Example: Using Google to find company details.

4. What is Footprinting through Social Networking Sites?

Attackers collect information from social media platforms.

Examples: Facebook, LinkedIn, Instagram

5. What is Competitive Intelligence?

Competitive intelligence means collecting information about competitors legally.

6. What is Social Engineering?

Social engineering is manipulating people to reveal confidential information.

Example: Pretending to be a technical support employee.

7. What are Footprinting Tools?

Footprinting tools help gather target information.

Examples:

- Whois
 - Maltego
 - Nmap
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8. What is Network Scanning?

Network scanning identifies active devices, open ports, and services in a network.

Purpose: Detecting vulnerabilities.

9. What is Port Scanning?

Port scanning checks open ports on a target system.

Example: Checking whether port 80 is open.

10. Name Some Port Scanning Tools

Tools:

- Nmap
 - Netcat
 - Angry IP Scanner
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11. What is TCP Scanning?

TCP scanning checks TCP ports to identify running services.

Advantage: Provides reliable results.

12. What is UDP Scanning?

UDP scanning identifies open UDP ports in a network.

Example: DNS services use UDP.

13. What is IDS?

IDS stands for Intrusion Detection System. It detects suspicious activities in networks.

14. What is Firewall Evasion?

Firewall evasion means bypassing firewall protection during scanning.

Example:

Using fragmented packets.

15. Difference Between Active and Passive Reconnaissance

Active Reconnaissance	Passive Reconnaissance
Direct interaction with target	No direct interaction
Easier to detect	Harder to detect
Faster information gathering	Safer information gathering

16. Difference Between TCP Scan and UDP Scan

TCP Scan	UDP Scan
Scans TCP ports	Scans UDP ports
More reliable	Slower
Connection-oriented	Connectionless

UNIT III – ENUMERATION AND VULNERABILITY ANALYSIS

1. What is Enumeration?

Enumeration is the process of extracting user names, network resources, and system information from a target system.

2. What is NetBIOS Enumeration?

NetBIOS enumeration collects shared resources and user details from Windows systems.

Example: Finding shared folders in a network.

3. What is SNMP Enumeration?

SNMP enumeration gathers information from network devices using SNMP protocol.

Example: Collecting router details.

4. What is LDAP Enumeration?

LDAP enumeration extracts information from directory services.

Example: User account details.

5. What is DNS Enumeration?

DNS enumeration collects DNS records and domain information.

Example: Finding website IP addresses.

6. What is SMTP Enumeration?

SMTP enumeration identifies valid email users from mail servers.

7. What is Vulnerability Assessment?

Vulnerability assessment identifies security weaknesses in systems and networks.

Purpose: To Improve system security.

8. What are OS Vulnerabilities?

OS vulnerabilities are weaknesses found in operating systems.

Example: Outdated software bugs.

9. What are Windows OS Vulnerabilities?

These are security weaknesses in Windows operating systems.

Examples:

- Weak passwords
 - Missing updates
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10. Name Tools for Identifying Windows Vulnerabilities

Tools:

- Nessus
 - OpenVAS
 - MBSA
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11. What are Linux OS Vulnerabilities?

Linux vulnerabilities are weaknesses found in Linux systems.

Examples:

- Misconfigured permissions
 - Old kernel versions
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12. What is Patch Management?

Patch management means updating software to fix vulnerabilities and bugs.

Benefit: Improves security.

13. What is Security Misconfiguration?

Improper security settings that create vulnerabilities.

Example: Using default passwords.

14. What is OpenVAS?

OpenVAS is an open-source vulnerability scanning tool.

Use: Detects security weaknesses.

15. Difference Between Vulnerability Assessment and Penetration Testing

Vulnerability Assessment	Penetration Testing
Finds vulnerabilities	Exploits vulnerabilities
Detection process	Attack simulation
Lower risk	Higher risk

16. Difference Between Windows and Linux Vulnerabilities

Windows Vulnerabilities	Linux Vulnerabilities
More malware attacks	Permission issues common
Closed-source OS	Open-source OS
Example: Registry attacks	Example: Kernel exploits

UNIT IV – SYSTEM HACKING

1. What is Web Server Hacking?

Web server hacking means attacking a web server to gain unauthorized access or steal information.

2. What are Web Application Components?

Web application components are the parts used in web applications.

Examples:

- Web server
 - Database
 - Browser
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3. What is SQL Injection?

SQL injection is a web attack where malicious SQL commands are inserted into input fields.

Purpose: Access or modify database data.

4. What is Cross Site Scripting (XSS)?

XSS is a web attack where attackers inject malicious scripts into websites.

5. What are Web Vulnerabilities?

Web vulnerabilities are weaknesses found in web applications.

Examples:

- SQL Injection
 - XSS
-

6. Name Some Web Security Testing Tools

Tools:

- Burp Suite
 - OWASP ZAP
 - Nikto
-

7. What is a Wireless Network?

A wireless network connects devices using radio waves instead of cables.

Example: Wi-Fi

8. Components of Wireless Network

Components:

- Router
- Access Point
- Wireless Adapter

9. What is Wardriving?

Wardriving means searching for wireless networks using a moving vehicle and wireless devices.

10. What is Wireless Hacking?

Wireless hacking means attacking wireless networks to gain unauthorized access.

11. What is WPA2?

WPA2 is a security protocol used to protect Wi-Fi networks.

12. What is MAC Filtering?

MAC filtering allows only authorized devices to connect to a wireless network.

13. Name Tools Used in Wireless Hacking

Tools:

- Aircrack-ng
 - Wireshark
 - Kismet
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14. What is Packet Sniffing?

Packet sniffing captures data packets moving through a network.

15. What is Wireshark?

Wireshark is a network packet analyzer used to monitor network traffic.

16. Difference Between SQL Injection and XSS

SQL Injection	XSS
Targets database	Targets browser
Uses SQL queries	Uses scripts
Steals database data	Steals cookies/session

17. Difference Between Wired and Wireless Network

Wired Network	Wireless Network
Uses cables	Uses radio waves
Faster and stable	Flexible and mobile
More secure	Easier to attack

UNIT V – NETWORK PROTECTION SYSTEMS

1. What is the Access Control List (ACL)?

ACL is a set of rules used to allow or deny network traffic.

Purpose: To Improve network security.

2. What is Firewall?

A firewall monitors and filters incoming and outgoing network traffic.

Purpose: It Blocks unauthorized access.

3. What is Cisco ASA Firewall?

Cisco ASA is a hardware firewall device used for protecting networks.

4. What are Firewall Configuration Tools?

These tools help configure and manage firewall settings.

Example: Cisco ASDM

5. What is Risk Analysis?

Risk analysis identifies possible threats and their impact on systems.

6. What is IDS?

IDS stands for Intrusion Detection System. It detects suspicious activities in networks.

7. What is IPS?

IPS stands for Intrusion Prevention System. It detects and blocks attacks automatically.

8. What is Network-Based IDS?

A network-based IDS monitors traffic across an entire network.

9. What is Host-Based IDS?

A host-based IDS monitors activities on a single computer.

10. What is Web Filtering?

Web filtering blocks harmful or unwanted websites.

Example: Blocking social media in offices.

11. What is a Security Incident Response Team?

This team handles and responds to cybersecurity incidents.

Functions:

- Detect attacks
 - Recover systems
 - Reduce damage
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12. What is Honeypot?

A honeypot is a fake system used to attract attackers.

Purpose: To Study attacker activities.

13. Advantages of Honeypots**Advantages:**

- Detect attackers
 - Collect attack information
 - Improve security
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14. What is Router Security?

Router security protects routers from unauthorized access.

Methods:

- Strong passwords
 - Firewall rules
 - Regular updates
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15. What is IPSec?

IPSec is a security protocol used to secure Internet communication.

Function:

Encrypts data packets.

16. Difference Between IDS and IPS

IDS	IPS
Detects attacks	Detects and blocks attacks
Monitoring only	Active protection
Sends alerts	Prevents attacks

17. Difference Between Network-Based IDS and Host-Based IDS

Network-Based IDS	Host-Based IDS
Monitors entire network	Monitors single system
Detects network attacks	Detects host attacks
Installed on network	Installed on device